



Power Aware Design

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CONTENTS

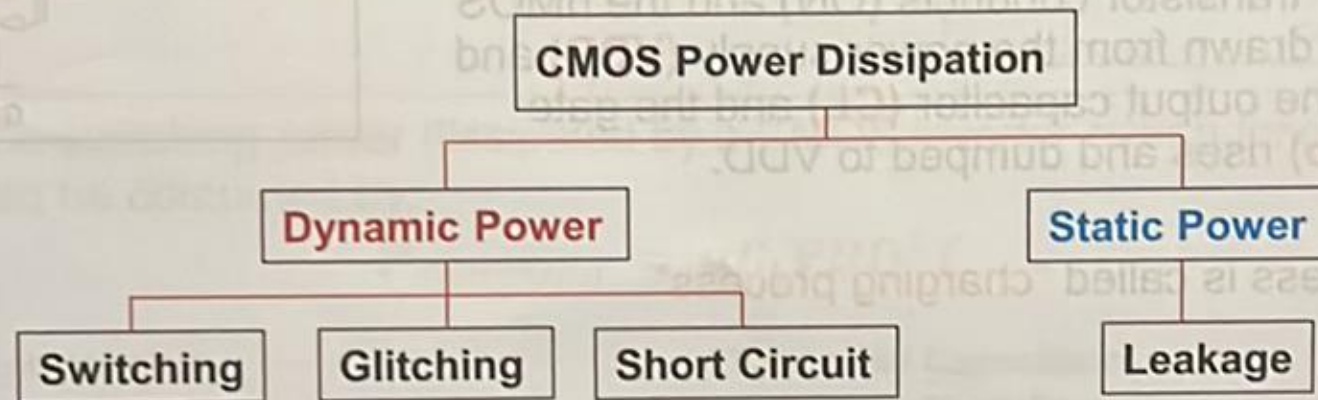
- **Types of Power Consumption**
 - **Switching Power**
 - **Short Circuit Power**
 - **Leakage Power**
 - **Glitch Power**
- **Power Reduction Techniques**
 - **Clock Gating**
 - **Multi Voltage Domains**
 - **Power Gating**
 - **Frequency Scaling**

CMOS POWER DISSIPATION



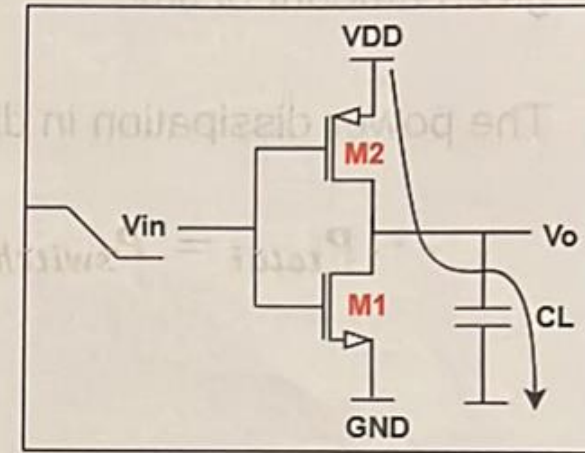
- Power dissipation is recognized as the most important parameter in modern VLSI design field, because the growth rate of the battery-based applications as wireless applications, laptops, and medical devices. Power consumption is directly proportional to the battery weight and volume required to operate circuits for a given amount of time.
- The power dissipation in digital CMOS circuits can be described by:

$$P_{total} = P_{switching} + P_{glitch} + P_{short-circuit} + P_{leakage}$$



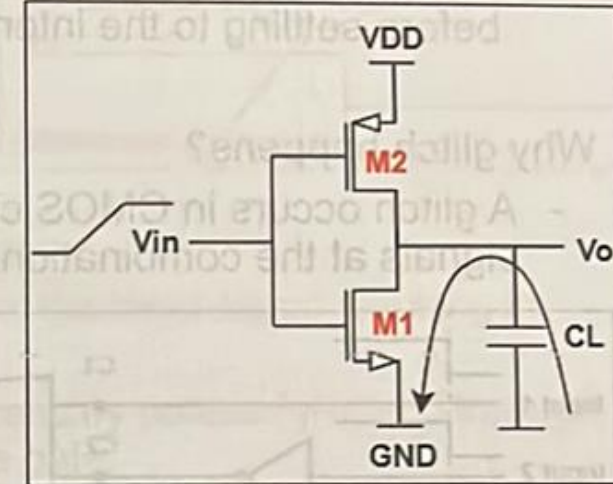
Switching Power

- The switching power $P_{\text{switching}}$ is caused by the charging and discharging of the load capacitances in the circuit when the circuit output switch from **high to low** and from **low to high** during the normal operation of the CMOS gates.
- Assume that the circuit is initially in a steady state, having as value '1' and output the logic value '0'. In this case the output capacitor is discharged.
- When the input waveform (**Vin**) undergoes a falling transition, transistor conducts (ON) and the nMOS transistor drawn from the power supply (**VDD**) and charges the output capacitor (**CL**) and the gate output (**Vo**) rises and dumped to VDD.
- This process is called "**charging process**"



Switching Power

- Assume that the circuit is initially in a steady state, having as input the logic value '0' and output the logic value '1'. In this case the output capacitor is charged.
- When the input waveform (**V_{in}**) undergoes a rising transition, transistor conducts (OFF) and the nMOS transistor current drawn from the directly from the output capacitor (**C_L**) to the ground (**GND**) and the gate output (**V_o**) falls and dumped to GND.
- This process is called “Discharging process”



- Then the switching power dissipated by a CMOS inverter over a time interval [0 T] can be computed by:

$$P_{\text{switching}} = \alpha C_L V_{DD}^2 f$$

V_{DD} : Supply Voltage

α : Activity Factor

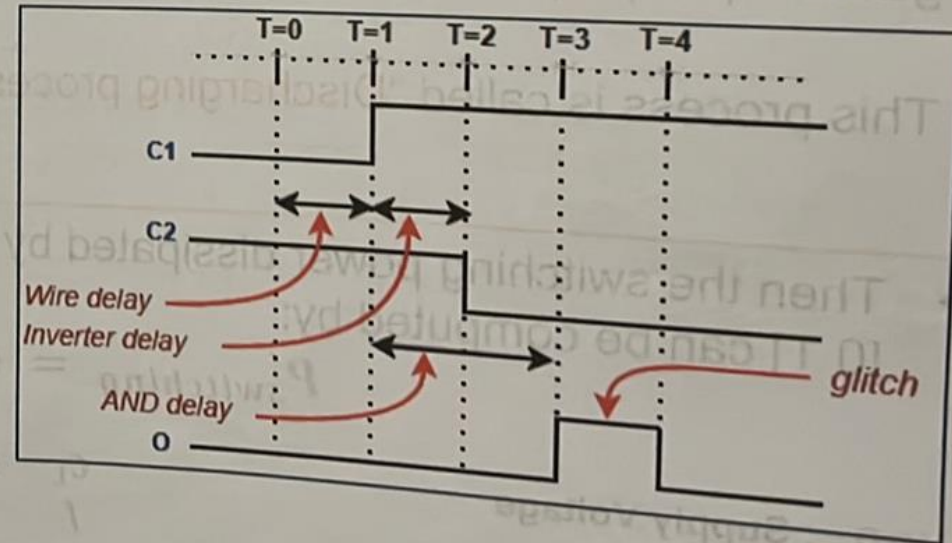
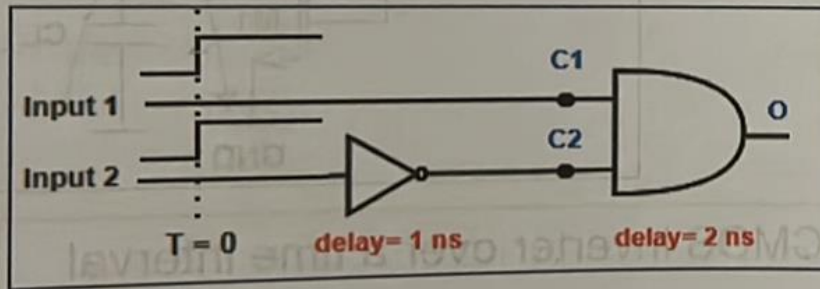
α = 1 (for clock signal) & **α = 0.5** (for rest of signals)

C_L : Total Load Capacitance

f : Operating Clock Frequency

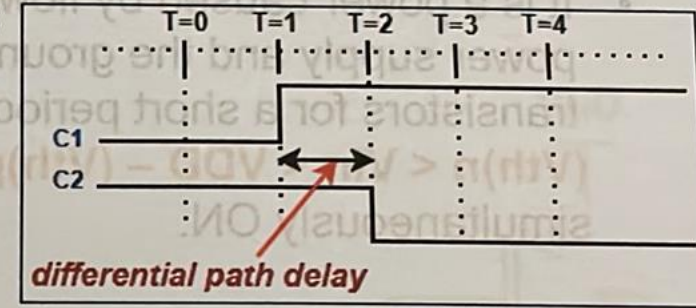
Glitch Power

- What is glitch?
 - A **glitch** is an unwanted pulse (small spike) at the output of a combinational gate that occurs when a single input change causes an output to change multiple times before settling to the intended value.
- Why glitch happens?
 - A **glitch** occurs in CMOS circuits due to the difference in the arrival times of the signals at the combinational gate inputs (inputs do not arrive simultaneously).



Glitch Power

- The maximum difference in the arrival time of the signals at the inputs of the gate is called **differential path delay**.



- Every combinational gate output has two types of transitions:-
 - Functional transition:** it is due to actual transmission of the input signal resulting in desired functioning of the logic gate.
 - Spurious transition:** it is due to transmission of unnecessary pulses (**glitches**) through the logic gate resulting in undesired functioning of the gate.
- Every signal of a gate needs to switch at most once in every clock cycle. But in the real scenario due to the glitches, there are output switching more than once in every clock cycle and these extra transitions raise the activity factor of a gate (α) above 1. So Glitches can lead to up to 40% of additional dynamic power consumption.

Short Circuit Power

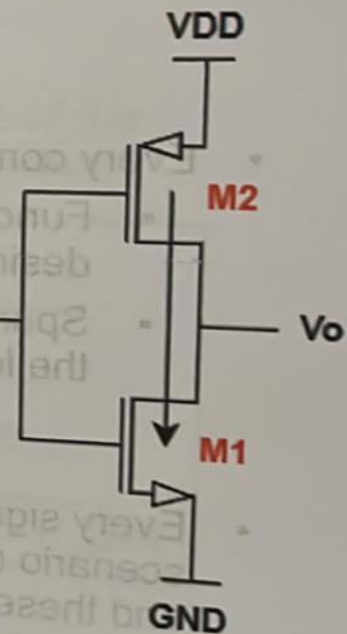
- It is a power caused by flowing a current through a **direct path** between the power supply and the ground during switching of both NMOS and PMOS transistors for a short period of time. It is happened when the condition **$(V_{th})_n < V_{in} < V_{DD} - (V_{th})_p$** is satisfied as both NMOS and PMOS are simultaneously ON.

- Condition for NMOS to be ON

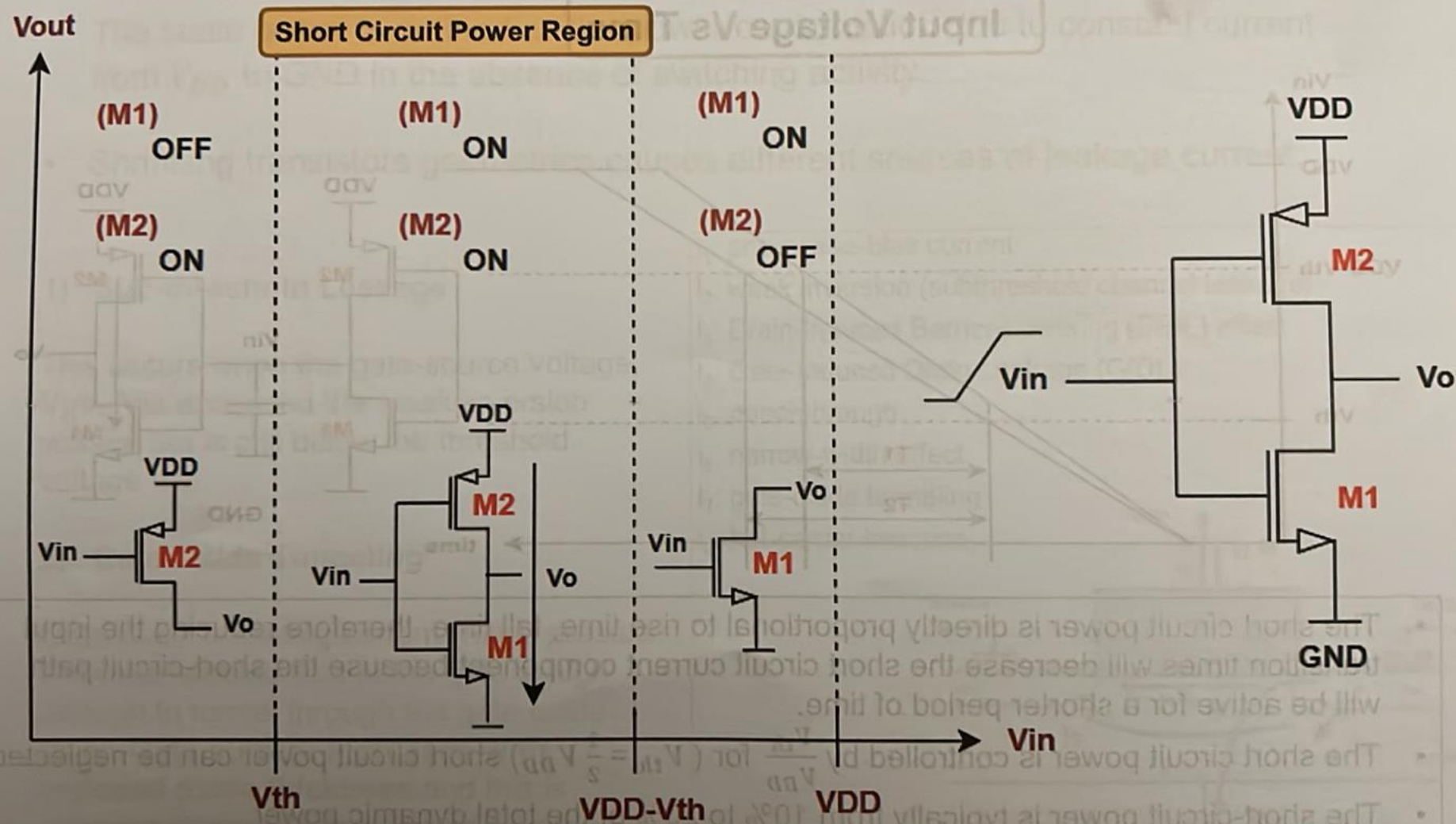
$$V_{gs} > V_{th} \Rightarrow V_{in} - 0 > V_{th} \Rightarrow V_{in} > V_{th}$$

- Condition for PMOS to be ON

$$V_{sg} > V_{th} \Rightarrow V_{DD} - V_{in} > V_{th} \Rightarrow V_{in} < V_{DD} - V_{th}$$

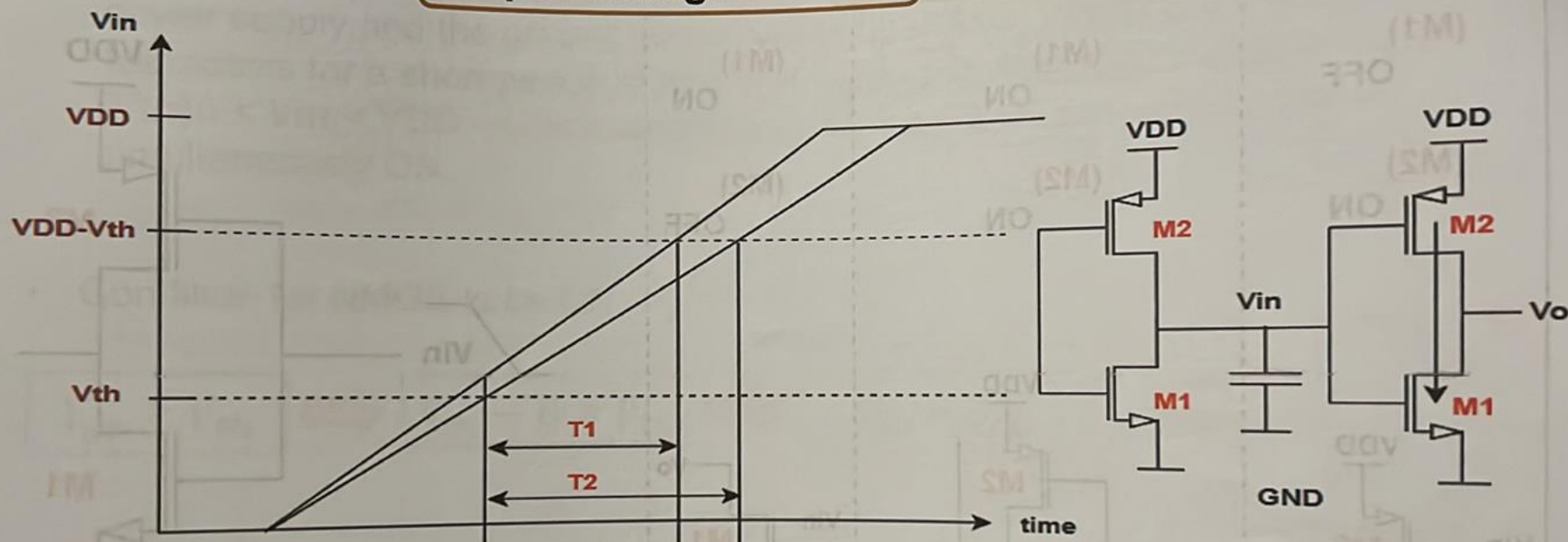


Short Circuit Power



Short Circuit Power

Input Voltage Vs Time



- The short circuit power is directly proportional to **rise time**, **fall time**, therefore reducing the input transition times will decrease the short circuit current component because the short-circuit path will be active for a shorter period of time. ☰
- The short circuit power is controlled by $\frac{V_{th}}{V_{DD}}$. for ($V_{th} = \frac{1}{2} V_{DD}$) short circuit power can be neglected
- The short-circuit power is typically from **10%** to **20%** of the total dynamic power. ☰

Leakage Power

- The static power is defined as the power consumption due to constant current from V_{DD} to GND in the absence of switching activity.
- Shrinking transistors geometrics causes different sources of leakage current.

1) Sub-threshold Leakage

This occurs when the gate-source voltage, V_{gs} , has exceeded the weak inversion voltage but is still below the threshold voltage V_{th} .

2) Gate Oxide Tunneling

Gate oxide tunneling current (I_G) is present when the electric field at the gate is high enough to tunnel through the gate oxide layer. **Gate current increases with reduced oxide thickness** and this is common during scaling down devices

I_1 : pn reverse-bias current

I_2 : weak inversion (subthreshold channel leakage)

I_3 : Drain-Induced Barrier-Lowering (DIBL) effect

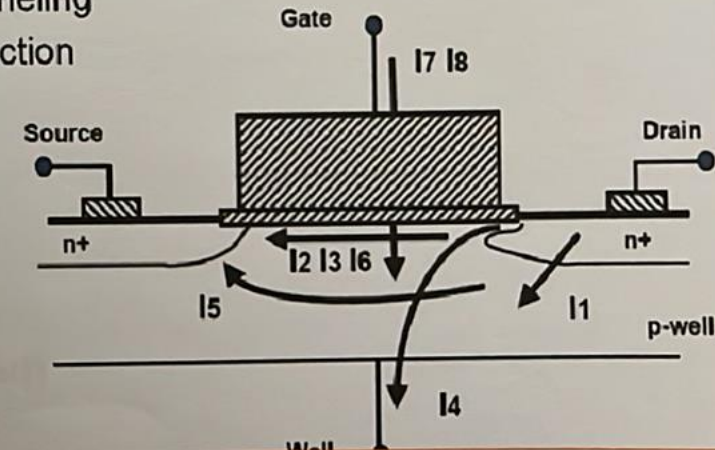
I_4 : Gate-Induced Drain Leakage (GIDL)

I_5 : punchthrough

I_6 : narrow-width effect

I_7 : gate-oxide tunneling

I_8 : hot-carrier injection



Power Reduction Techniques



- **Why Low Power ?**

- We want to add more features to the devices but we want to reduce also the size (size is limited by the battery)
- we want to extend the lifetime (also limited by the battery)

- **Factors Affecting Dynamic Power**

- Switching activity (α)
- Frequency
- Supply Voltage
- Output Node Capacitance

$$P_{switching} = \alpha C_L V_{DD}^2 f$$

- **Dynamic Power Reduction Techniques**

- To reduce Switching activity (α)
 1. **Clock Gating (Front End)**
 2. **Avoid Comparison (Front End)**
 3. **Operand Isolation (Front End)**
 4. **Shifter Instead of Multipliers (Front End)**

Power Reduction Techniques

- **Dynamic Power Reduction Techniques**

- To reduce Operation Frequency
 - **Multiple Frequencies (Front End)**
- To reduce Supply Voltage
 - **Multiple Voltage Domains (Back End)**
- To reduce Gate Capacitance
 - **Small Gate Size & Fewer Stages of Logic**
- To reduce Wire Capacitance
 - **Good Floorplanning to keep communication blocks close to each others**

- **Short Circuit Power Reduction Techniques**

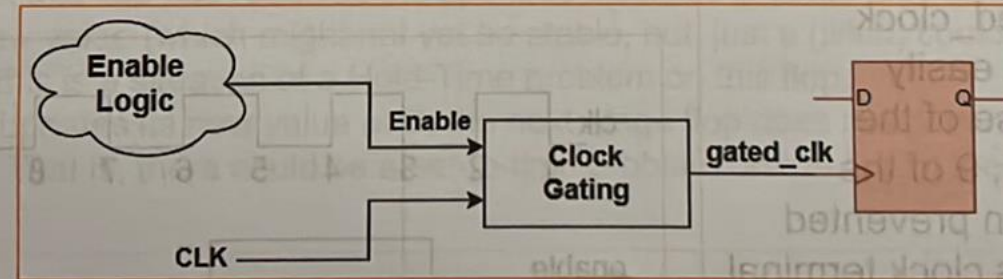
- **Multi Threshold (MVT) Voltage (Back End)**

- **Leakage Power Reduction Techniques**

- **Multi Threshold (MVT) Voltage (Back End)**
- **Power Gating (Back End)**

Clock Gating

- It is the most popular and most effective method to reduce **dynamic power** consumption.
- Dynamic power is consumed in a circuit when a flop in the design switches state due to a clock trigger due to :-
 - power consumed by the combinational logic when input values are changing on each clock.
 - power consumed by the flip flop
 - Power consumed by the clock tree buffer.
- Some blocks may not be getting used at certain time. So we can reduce dynamic power by turning off the clock to registers in unused blocks to save **clock activity ($\alpha = 1$)** and eliminate **all switching activity in this block**. This is achieved by having **enable logic** before the clock and such a clock is called **gated clock**.



- There are two types of clock gating styles :-
 - Latch-free clock gating
 - Latch-based clock gating

Clock Gating

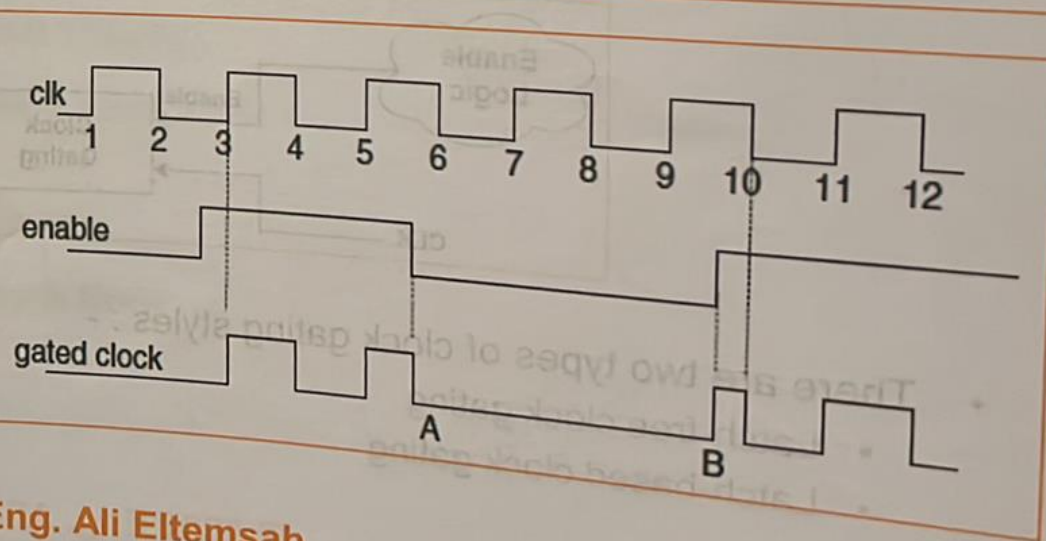
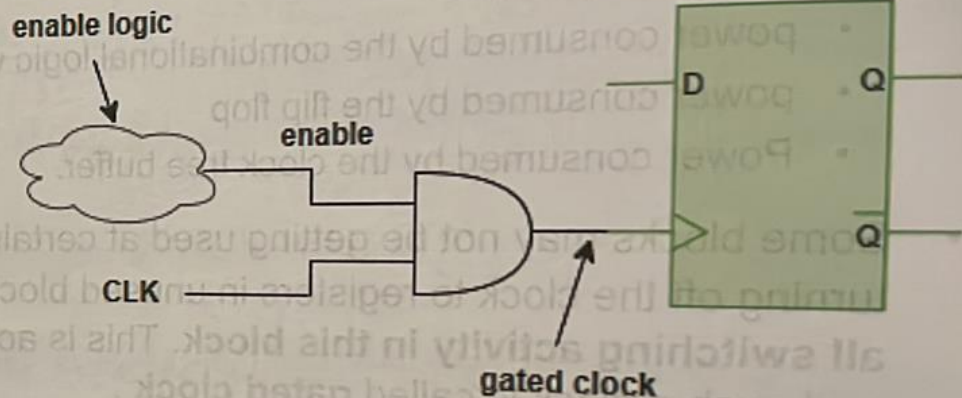
Latch-free Clock Gating

Implementation

- The Latch-free Clock Gating circuit is simply an **AND Gate** which enables or disables the clock to cross the gate and reach the flop's clock terminal.

Waveform

- We can notice the gated_clock waveform. It should be easily observable that because of the gating, edges 1, 2, 7, 8, 9 of the original clock have been prevented from reaching the flop's clock terminal due to disable enable signal, thus saving power within the flop.



Clock Gating

Latch-free Clock Gating Problems

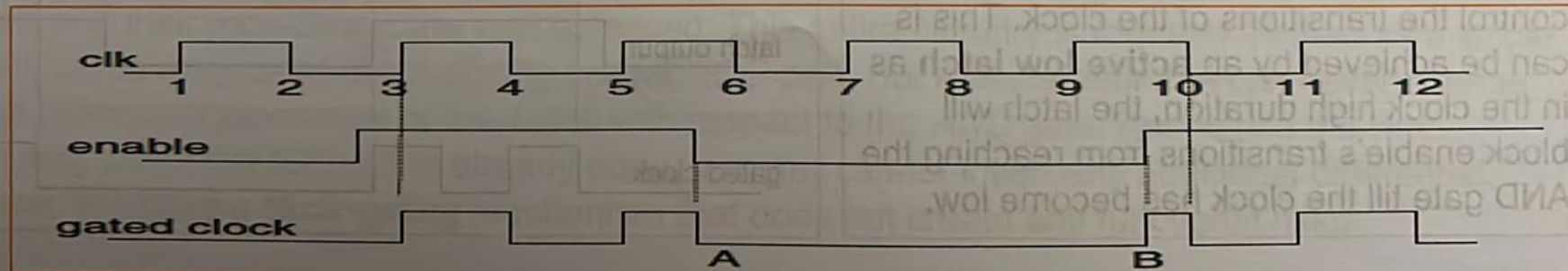
1. Pulse Clipping

If we look closely at the point A on the waveform for the gated circuit. The falling edge which should have come at 6 has come earlier. Effectively, the High pulse of the clock has been shortened. This is called “pulse clipping” and can result in **Pulse width Violation** for the flop being clocked by this gated-clock. 📄

2. Spurious Clocking

If we look at the point B on the waveform for the gated circuit. There is a positive edge at point B on the gated clock. This is not where a clock should have come. This is called “**spurious clocking**”. This can result in two possible problems:-

- By the time that this flop has received the spurious clock, its D could also be getting updated with the new values. This new value (which might not yet be stable, but, just a glitch) could get captured in the flop. In terms of STA, this is a situation of a **Hold-Time problem on this flop**.
- Since this flop updates its new value at B, the next stage flop does not get one complete cycle for the value to reach there. That is, there could be a **setup-time problem on the flop at the next stage**.

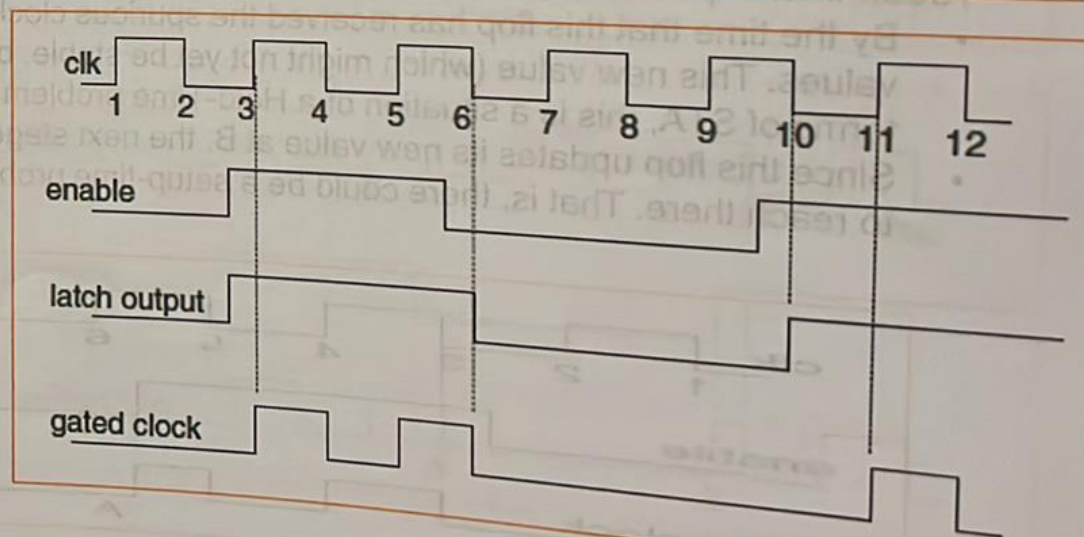
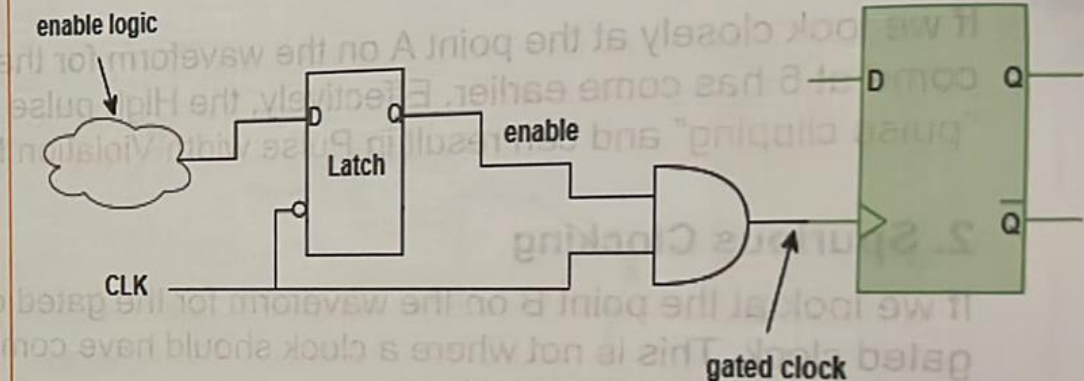


Clock Gating

Latch-based Clock Gating

Implementation

- The Latch-based Clock Gating circuit is composed of **Active Low Latch** controls the transmission of enable signal to the **AND Gate** which enables or disables the clock to cross the gate and reach the flop's clock terminal.
- To prevent **Pulse-Clipping** or **Spurious Clocking**, you need to ensure that the enable signal is received on the gate only when the clock is Low. That way, the enable itself will not cause any transition at the output of the gate, but it will be able to control the transitions of the clock. This is can be achieved by an **active low latch** as in the clock high duration, the latch will block enable's transitions from reaching the AND gate till the clock has become low.



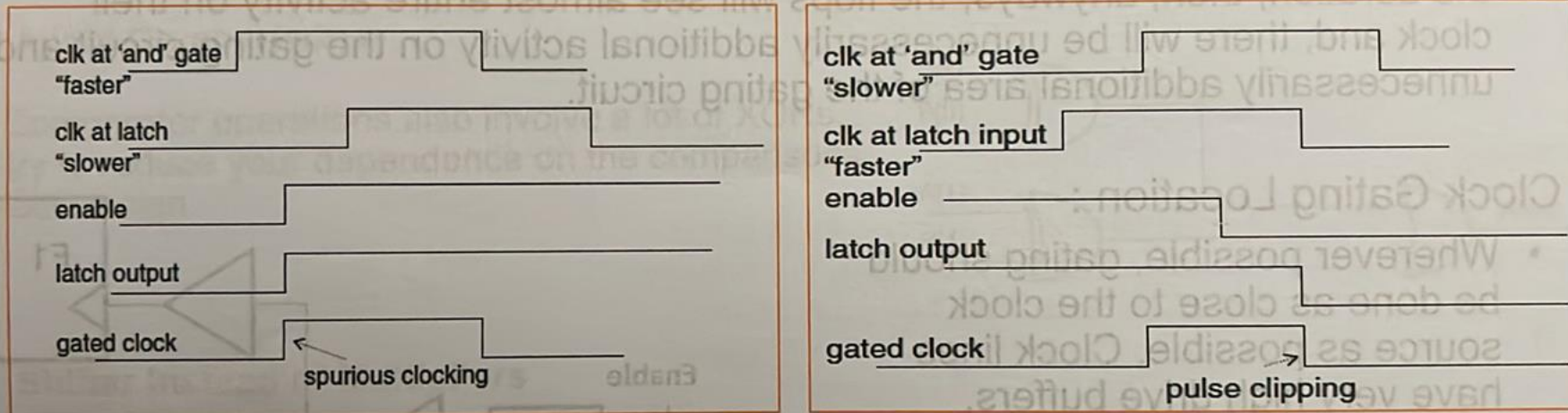
Clock Gating

Clock Gating



ICG (Integrated Clock Gating) Cell

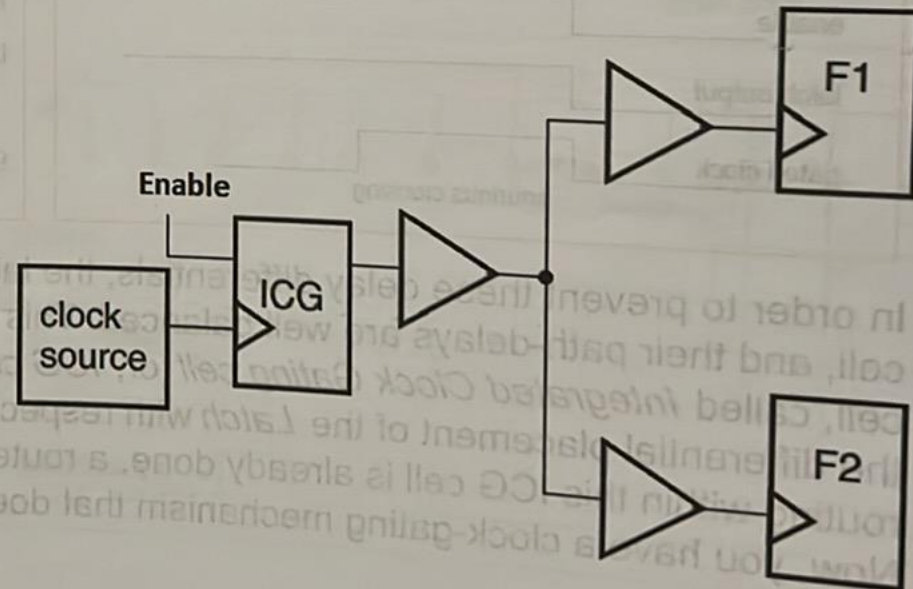
- Due to differential routing delays, it so happens that the clock to the AND gate path is faster or slower than the clock-to-latch path, which allow spurious clocking and pulse clipping to be caused again as shown in the following waveforms.



- In order to prevent these delay differentials, the latch and the AND gate are put together in a cell, and their path-delays are well balanced. This entire circuitry is together treated as a single cell, called *Integrated Clock Gating cell* (or, *ICG cell*). So, a placement tool can never change the differential placement of the *Latch* with respect to the *AND* gate. And, since the wire-routing within this ICG cell is already done, a router cannot insert any additional differential. Now, you have a clock-gating mechanism that does not create any functional risks

Clock Gating

- Clock Gating Recommendation :-
 - Clock gating makes sense only if the enable signal is expected to be OFF for a significant duration. so that a significant number of clock pulses can be eaten up at the clock gating logic. Otherwise, if the gate is expected to be ON for most of the duration, then, anyways, the flops will see almost entire activity on their clock and, there will be unnecessarily additional activity on the gating circuit and unnecessarily additional area of the gating circuit. ☹️
- Clock Gating Location :-
 - Wherever possible, gating should be done as close to the clock source as possible. Clock lines have very high drive buffers, which consume a lot of power. So, the closer to the source you do the gating, the more power you can save. ☹️

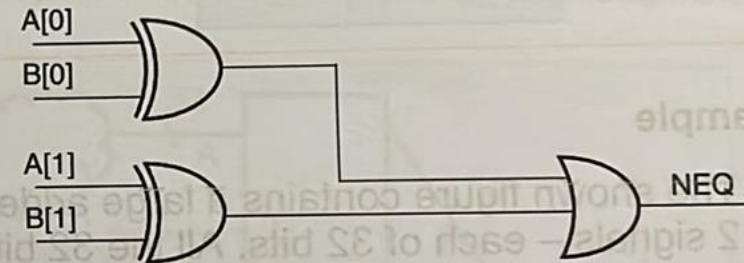


Avoid XOR Gates

- In an XOR gate, any transition at any of the inputs will always cause a transition at the output. Compare that with gates like AND, NAND, OR, NOR etc. Not all the transitions at an input will result in a transition at the output. At least, some of the transitions will not reach the output, because the values get blocked by the other input (0 for AND, NAND and 1 for OR, NOR). So, in general, an XOR gate is more power hungry, because it transmits all transitions at inputs into a transition at output.

1) Avoid Comparisons

- Comparator operations also involve a lot of XORs. So try to reduce your dependence on the comparisons in your design.

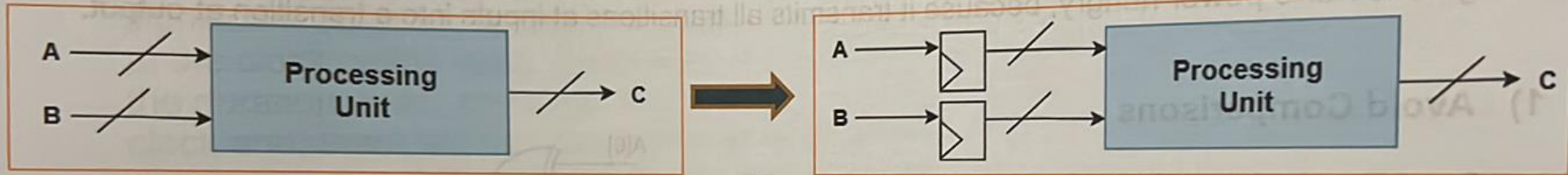


2) Shifter Instead of Multipliers

- Multipliers (and adders) are made of a large number of XORs. So, even a single transition at any input will result in a lot of transitions internal to the multipliers. There are certain multiplication values, where, the same result can be achieved using another operation, e.g. shifter instead of multiply by 2, 4, 8 etc. Where possible, the multiplier should be replaced by a Left Shifter. Similarly, some divisions (by 2, 4, 8 etc.) can be replaced by a Right Shifter.

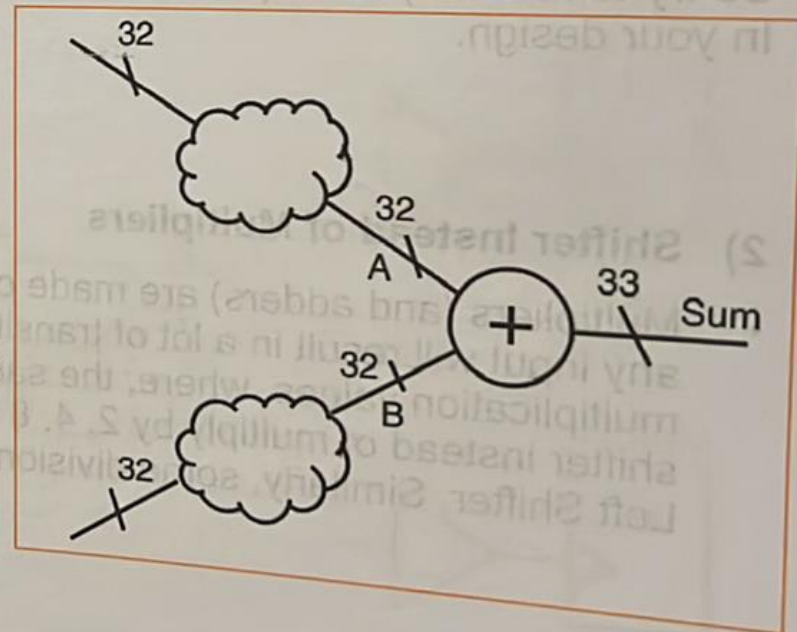
Operand Isolation

- In case of you are doing some processing on a data buses as A and B, you have to **isolate or register the inputs to avoid unnecessary glitching power** due different arriving times of the bits of the data buses due to different routing tracks.



Example

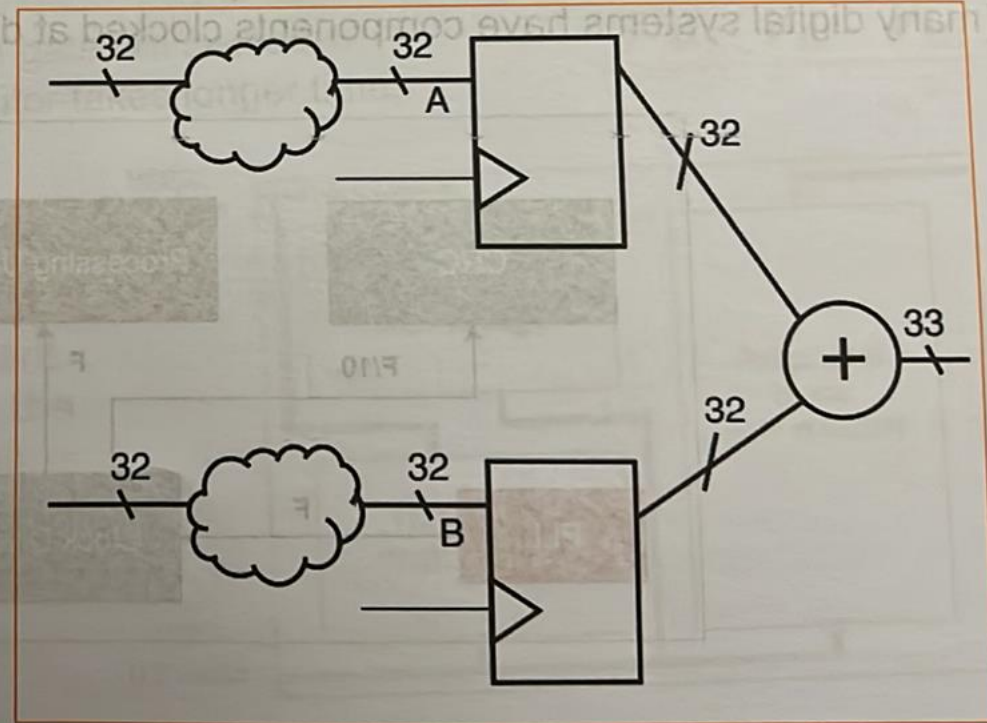
- The shown figure contains a large adder, which adds 2 signals – each of 32 bits. All the 32 bits of the busses A or B will not arrive at the adder at the same instant. These different bits could be arriving through different combinational gates, or, through different routing tracks. As each signal at the input of the adder changes, the adder reevaluates and the output switches. This activity starts with the arrival of the earliest of the 64 bits (32 bits each of each A and B), and, continues till all the bits have arrived. However, only the final addition value is of interest. All the intermediate results were glitches and not used, but, they consumed power



Operand Isolation

- In general Arithmetic operators such as Adders and Multipliers have lots of XORs. So, any transition at the inputs will reach the output – thus causing more switching activity down the circuit. So, the aim is to reduce the number of activities at the inputs of these operators. This can be done by latching all the inputs, and, then, feeding the latch output directly to the adder. This ensures that all the bits are presented to the operator at almost the same time. This prevents the operator to evaluate multiple times. Hence, a lot of spurious switching can be avoided. Since all the operands are now coming directly from the registers, they will reach the operator at almost the same time.

- The cost of using this technique: -
 - Add Latency 1 Clock Cycle.
 - Additional Silicon Area

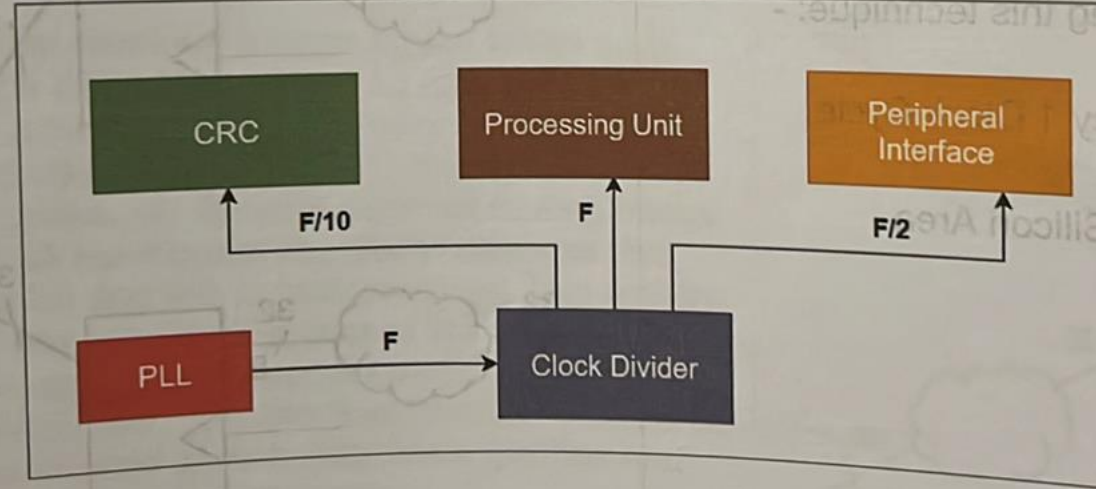




Multiple Frequencies Technique

- Multi frequencies technique is used for reducing the **dynamic** power of the design.
- Dynamic power is directly proportional to the operating frequency.
hence reducing operating frequency improves the power performance.
- If a module's clock frequency can be reduced while **maintaining acceptable performance** and **correct functionality**, its frequency can be reduced to reduce total power dissipation, so many digital systems have components clocked at different frequencies.

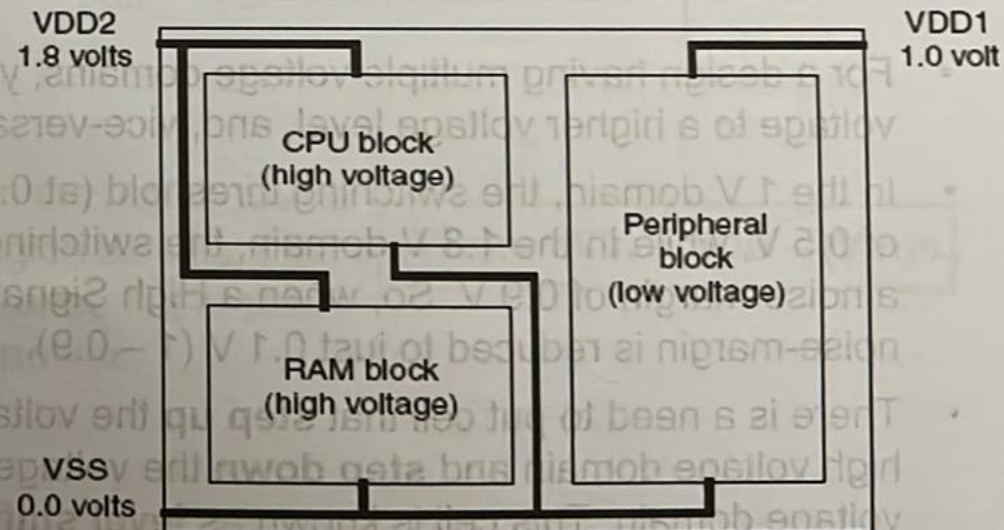
$$P_{dynamic} \propto F$$



Multiple Voltage Domains

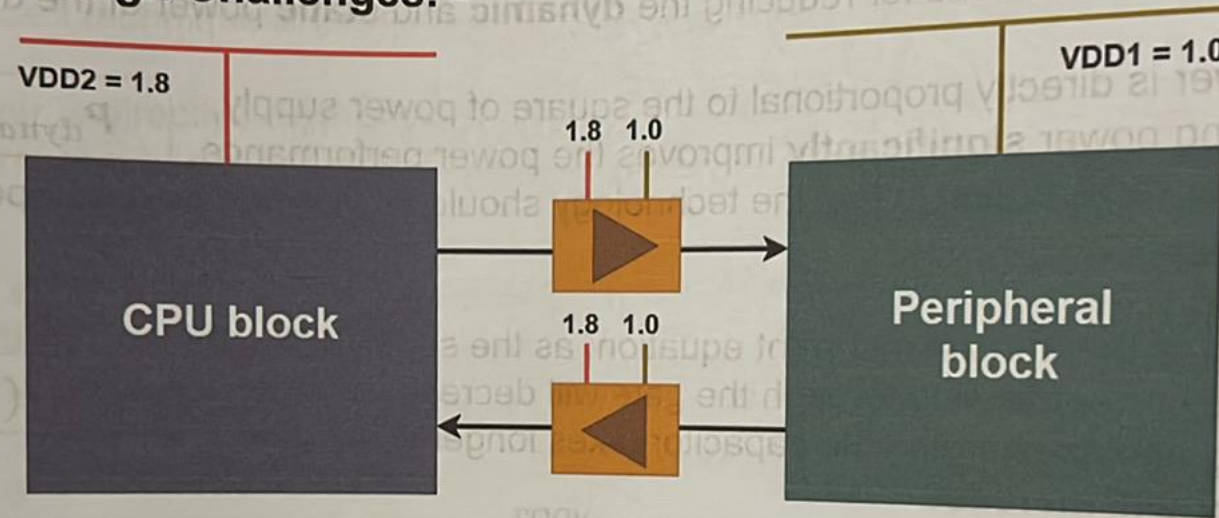
- Multi VDD technique is used for reducing the **dynamic** and **static** power of the design.
- Dynamic power is directly proportional to the square of power supply, hence reducing power **significantly** improves the power performance. The only major requirement is that the technology should support the devices operating at a different voltages.
$$P_{dynamic} \propto VDD^2$$
- But according to the transistor current equation, as the supply voltage decrease, the current flowing through the gate will decrease. then charging and discharging the capacitor takes longer time, so **gate delay will increase**
$$I \propto (V_{in} - V_{th})$$

- To avoid system performance degradation, we can provide different voltage to different blocks. **high voltage can be applied to the timing critical blocks** as processing unit, while lower voltage used for blocks with enough timing slack. By this way Overall system performance is maintained.



Multiple Voltage Domains

Multi Voltage Design Challenges: -



- For a design having multiple voltage domains, you would have signals going from a lower voltage to a higher voltage level, and, vice-versa. This can have an impact on noise-margin.
- In the 1 V domain, the switching threshold (at 0.5 Vdd) is 0.5 V, which means a noise-margin of 0.5 V, while In the 1.8 V domain, the switching threshold (at 0.5 Vdd) is 0.9 V, which means a noise-margin of 0.9 V. So, when a High Signal at 1 V signal reaches the 1.8 V domain, the noise-margin is reduced to just 0.1 V ($1 - 0.9$).
- There is a need to put cell that **step up** the voltage when moves from lower voltage domain to high voltage domain and **step down** the voltage when moves from high voltage domain to low voltage domain. This cell is known as **level shifter**

Multi Threshold (MVT) Voltage

- Multiple threshold voltage technique is used for reducing **short circuit** and **leakage** power.
- Multiple threshold voltage techniques use both **Low V_{th}** and **High V_{th}** cells. The only requirement is that the **technology library** should support devices operating at **different Threshold Voltage**.
- In order to obtain transistors with different threshold voltages, during the manufacturing process, several characteristics of the transistors may vary, such as channel doping, channel length, and the thicknesses of gate oxide.

- One of the main leakage power components is the sub threshold current which varies exponentially with threshold voltage. **Higher V_{th} results in lower leakage power**

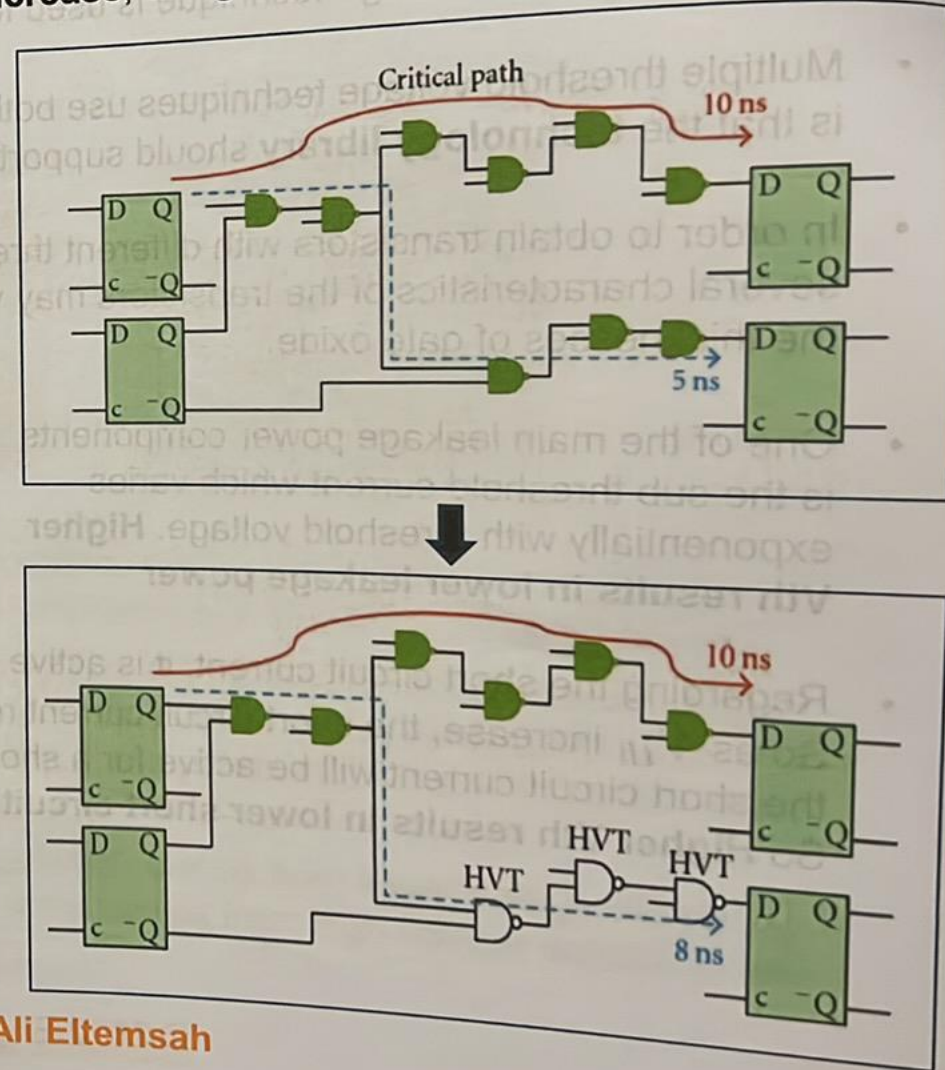
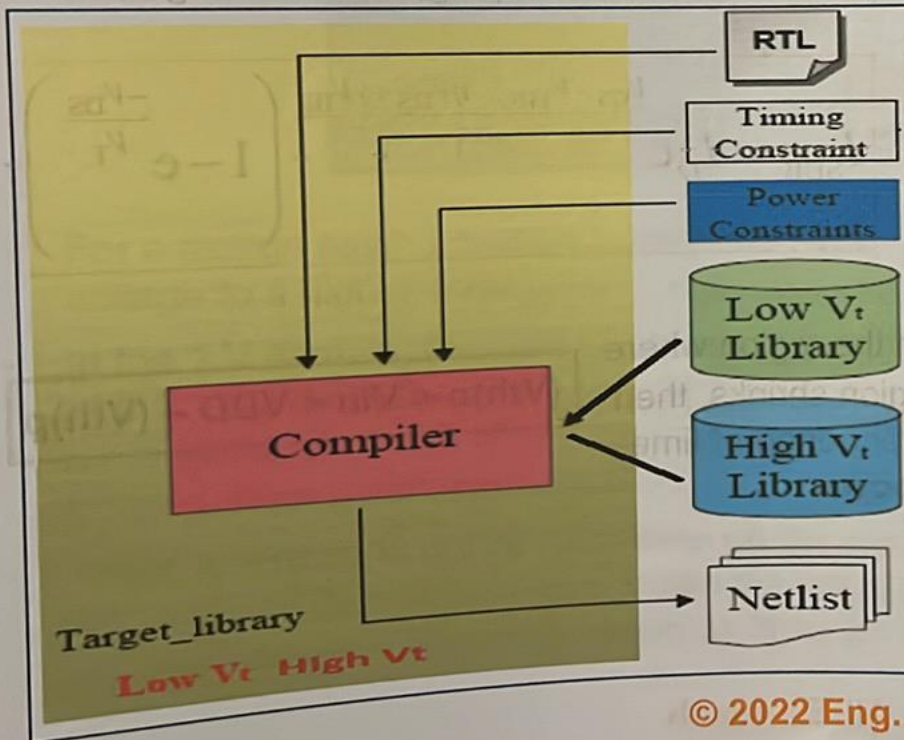
$$I_{SUB} = I_O e^{\frac{V_{GS} - V_{THO} - \eta V_{DS} + \gamma V_{BS}}{nV_T}} \left(1 - e^{\frac{-V_{DS}}{V_T}} \right)$$

- Regarding the short circuit current, it is active in the region where So as V_{th} increase, the short circuit current region shrinks, then the short circuit current will be active for a shorter period of time. So **Higher V_{th} results in lower short circuit power.**

$$(V_{th})_n < V_{in} < V_{DD} - (V_{th})_p$$

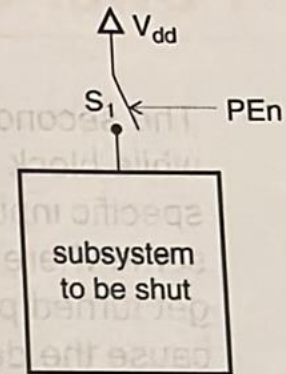
Multi Threshold (MVT) Voltage

- According to Mosfet Current Equation, as V_{th} increase, the gate delay will increase as the flowing current through transistor will decrease.
- Good design practice starts with **high-Vt** Cells everywhere and selectively introduces **low-Vt** devices in the critical paths.

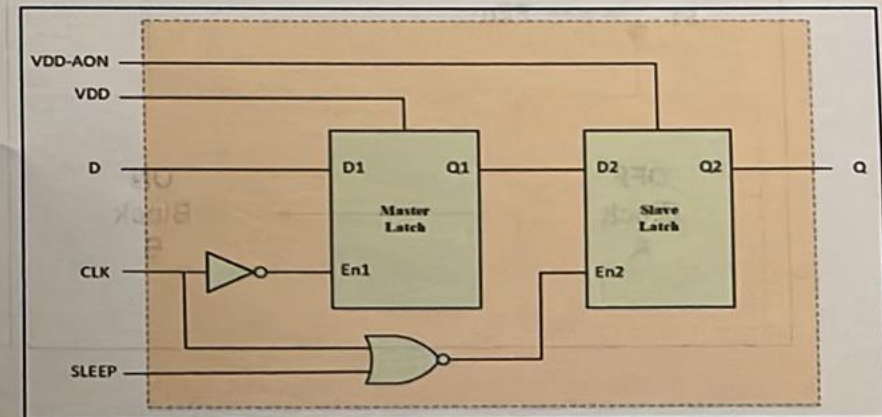
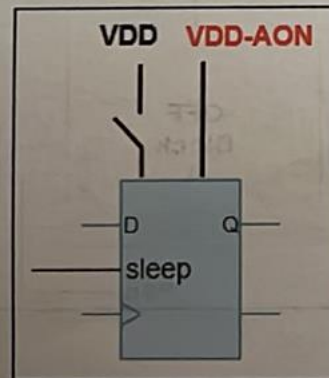


Power Gating

- Power Gating technique is used for reducing only **leakage** power of the design.
- If a portion of the circuit which are not in use (IDLE) for long durations. The power supply to this portion of the circuit might be switched off, the switch S_1 is usually an active device (ex: PMOS) controlled through **PEn (Power Enable)** signal. 
- Power Domain refers to a portion of the circuit, which operates at the same voltage level and turns ON or OFF together.
- The most important aspect is that when the system receives power back (to start an activity), it should be able to start from where it was when the power was shut off. Hence, all the sequential devices should be able to retain their stored values, even when they were powered down. Such flops are called **Retention Flops**. These special flops have multiple power supplies to allow the flops to be always powered up even when **PEn** is down. 



Retention Flop



Power Gating

- The second block **B** could be ON, because it does not have a Power Switch (i.e., its Always ON) while block **B** is expecting a signal from a block **A** that has been turned OFF. This could cause this specific input of B to be floating. A floating signal might neither be at 0 or at 1; rather it might lie somewhere in-between the two levels. Thus, inside block B, both the P and the N transistors might get turned partially ON – thus, causing a direct current path from supply to the ground. This would cause the device to be burned. Thus, there is a need for an **Isolation Cell**.
- The Isolation Cell sends the normal signal, when Powered ($P_{en} = 1$), and a 0 during Power Down ($P_{en} = 0$).

